

Andreas Hovaldt HØJRUP

MSc in Computer Engineering, specialized in AI, vision and sound
BSc in Robotics

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I recently graduated as a computer engineer, specialized in AI, vision, and sound. I finished my MSc in Computer Engineering in June 2026, with a previous bachelor's in Robotics. My work and university experience spans across a broad range of the ML stack, from applying 2D and 3D visual foundation models to in-field LiDAR perception in my AGCO master's thesis, to approximating model predictive controllers during a five-month internship at CERN, down to the GPU-cluster training, experiment tracking, and Linux infrastructure underneath. My native language is Danish, but throughout my work and studies, I have also become fluent in English. I am passionate about modern technology and building things for real world problems. I am now seeking ML engineering, computer vision, or MLOps roles where I can solve real problems, contribute to exciting projects, and keep learning.

📁 WORK EXPERIENCE

Jun 2026 Feb 2026	Student Worker Applied Vision Research, AGCO A/S, Randers, Denmark (hybrid) ➢ Trained lightweight task adapters on frozen DINOv3 embeddings to predict machine and material state from in-machine cameras ➢ Designed a per-instance gating loss that lifted task accuracy from ~70% to over 99% ➢ Scaled the dataset from 41k to 158k images for multi-target regression ➢ Implemented sequence models which beat single-frame (R^2 0.31 $\rightarrow$ 0.63) ➢ Built gradient-rollout explainability tooling PyTorch DINOv3 Vision Foundation Models Weights & Biases AI Explainability
Jan 2026 Sep 2025	Computer Engineering Intern, CERN, Geneva, Switzerland ➢ Built neural-network approximation of an MPC for HVAC air-handling units, cutting online compute by ~460× ➢ Developed the full pipeline : Apache Spark data extraction, dataset curation, modular PyTorch training, and closed-loop simulation deployment ➢ Compared MLP, TCN, and transformer models, best reached $R^2 = 0.995$ (regression) and 94% class-averaged accuracy (classification) ➢ Held zone temperatures within operational bounds in closed-loop run within the EcosimPro digital twin simulation PyTorch ONNX Runtime Transformer Time series Apache Spark (NXCALS) Weights & Biases EcosimPro
Aug 2025 Jul 2025	Student Worker 3D Perception Research, AGCO A/S, Randers, Denmark ➢ Researched and evaluated point-cloud segmentation techniques for agricultural environments, laying groundwork for the thesis collaboration Point Cloud Segmentation Open3D Pointcept
Nov 2023 Feb 2019	Postal Worker & Assistant Coordinator, POSTNORD, North Jutland, Denmark ➢ Full-time during a sabbatical year, then part-time alongside studies ➢ Assistant coordinator for weekend parcel delivery across North Jutland municipalities

🎓 EDUCATION

2024-2026	MSc in Computer Engineering with a specialization in AI, vision, and sound at Aalborg University Overall average : 11.3 Thesis : 12 (Danish 7-point scale)
2021-2024	BSc in Robotics at Aalborg University Overall average : 10.0 Thesis : 10 (Danish 7-point scale)

🔬 PUBLICATION

Synthetic LiDAR Data Generation using 3D Gaussian Splatting for 3D Semantic Segmentation Jun 2025
Christiansen, A.T.* , Højrup, A.H.* , Stephansen, M.K.* , et al. (*Equal contribution)
Synthetic Data for Computer Vision Workshop, CVPR
[arXiv:2506.05009](https://arxiv.org/abs/2506.05009) | [OpenReview](#)

PROJECTS

Foundation Models in Agricultural 3D Perception (MSc Thesis, with AGCO)

Feb 2026 - Jun 2026

Evaluated 2D and 3D visual foundation models for in-field LiDAR detection and segmentation, on a self-captured, manually annotated agricultural LiDAR dataset alongside ScanNet v2. Domain fine-tuning a 3D point-cloud foundation model raised detection mAP25 by $\sim 4\times$, while injecting frozen DINOv3 image embeddings lifted trained from-scratch models by $\sim 8.3\times$.

PyTorch 3DETR DINOv3 LiDAR Pointcept Slurm Transformer Detection Segmentation

Visual Servoing for Vertical Farming (BSc Thesis, with Seasony)

Feb 2024 - Jun 2024

Compared classical and CNN-based visual servoing for aligning the Watney C robot to racks in a vertical-farming setup. The CNN approach performed best, meeting the error-margin requirement 99.8% of the time on the hardest alignment task (± 8.5 mm on the global Z-axis).

CNN Visual Servoing Computer Vision Robotics NVIDIA Isaac Sim

Bimanual Manipulation with an Integrated LLM

Sep 2023 - Jan 2024

Natural-language-controlled bimanual system using two Kuka IIWA 7 arms, with YOLOv8 object detection and a GPT-4 agent translating prompts into robot commands. Achieved a 73% task success rate while demonstrating LLM-driven adaptive task execution.

LLM Agents YOLOv8 GPT-4 ROS2 Robotics LangChain

Position Control of a Crazyflie 2.1 Drone via Vicon

Feb 2023 - Jun 2023

Cascading control loop for x/y/z position control of a Crazyflie 2.1 micro-drone using a Vicon motion-capture system to execute pre-planned trajectories. Steady-state error of ~ 2 mm (x/y) and ~ 7 mm (z), with 4.2% average overshoot.

Control Systems Motion Capture Trajectory Planning

PERSONAL PROJECTS

Self-Hosted Homelab

Self-hosted home server running 10+ containerized services (Docker/LXC) on Proxmox VE, including reverse-proxied web apps, automated media management, game servers, and monitoring. Secured with a WireGuard VPN and Cloudflare mutual-TLS, with internal services kept behind authenticated access.

Proxmox VE Docker LXC WireGuard Cloudflare Linux

Personal Finance Tracker

Full-stack web app that fetches and categorizes bank transactions through the GoCardless API, with a FastAPI + PostgreSQL backend and a React frontend, containerized with Docker Compose and self-hosted.

FastAPI PostgreSQL React GoCardless API Docker Compose

TECHNICAL SKILLS

Programming	Python (advanced); MATLAB, C++, C# (basic)
ML & Vision	PyTorch, scikit-learn, ONNX Runtime, OpenCV, Open3D, Pointcept, Weights & Biases, visual foundation models (DINOv3, Utonia), 3D Gaussian Splatting
Data & Compute	NumPy, SciPy, Pandas, Apache Spark, Slurm GPU clusters (HPC)
Infrastructure	Linux, Docker, Proxmox VE, Git, FastAPI, NGINX, WireGuard
Robotics	ROS2, NVIDIA Isaac Sim, sensor integration (LiDAR, RGB-D cameras)

INTERESTS

Outside of work, I still like to sometimes have something hands-on to tinker with. Previously I have worked on small home electronics projects with either an Arduino or a Raspberry Pi. Recently, I have focused on tinkering with my self-hosted Proxmox setup. When I'm not working on something, I like to go for my weekly runs or, in the weekends, go hiking, and a pleasant evening with friends is always a nice way to unwind.

REFERENCES

Available upon request.